# MSHA Block 1a Ontology Refinement

## Block 1 Versus Block 1a Development Comparison

Generated October 15, 2026 at 05:29

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Model: Qwen/Qwen2.5-7B-Instruct
Prompt version: MSHA_BLOCK1A_ONTOLOGY_REFINEMENT_V2.0
Development records reprocessed: 1,000
Block 1 successful results: 991
Block 1a successful results: 945
Block 1a review-required records: 562

Purpose:
Evaluate whether stricter category definitions, improved routing,
evidence requirements, and factor-based prevention mapping reduce
category drift before the ontology is frozen for Block 2.
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# Executive Summary

Block 1a reprocessed the same development records used in Block 1. No new development sample was selected.

The refinement tightened the definitions of unstable material, housekeeping, securement, line of fire, work method, and no-separate-factor cases.

The cause ontology was expanded to identify foreign-body or particle contact and acute heat or environmental exposure. Acute strains and hernias were explicitly included under manual handling or overexertion.

The prevention theme is now selected from the contributing factor using a preferred factor-to-prevention mapping. When no separate factor is stated, the system uses INVESTIGATION_REQUIRED rather than inventing training, PPE, material securement, or procedure improvements.

Block 1 review-required rate: 60.8%
Block 1a review-required rate: 56.2%

Block 1 factor-prevention alignment rate: 46.0%
Block 1a factor-prevention alignment rate: 81.0%

The remaining sections show where labels changed and whether the revised system reduced the category concentrations observed in Block 1.

# Block 1a Revisions

1. Added record-type classification within the model output.

2. Added FOREIGN_BODY_OR_PARTICLE_CONTACT.

3. Added HEAT_OR_ENVIRONMENTAL_EXPOSURE.

4. Explicitly routed acute hernias and strains to
   MANUAL_HANDLING_OR_OVEREXERTION when supported.

5. Restricted POOR_HOUSEKEEPING_OR_SURFACE_CONDITION to explicit
   walking, standing, access, or working-surface hazards.

6. Restricted UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL to explicitly
   unstable, loose, unsupported, rolling, shifting, or falling material.

7. Activated INADEQUATE_SECUREMENT_OR_RESTRAINT when missing or failed
   blocking, fastening, support, restraint, or securement is stated.

8. Emphasized LINE_OF_FIRE_BODY_POSITION for pinch points and exposure
   to moving or falling objects.

9. Replaced generic NOT_STATED with NO_SEPARATE_FACTOR_STATED.

10. Required NO_SEPARATE_FACTOR_STATED to map to
    INVESTIGATION_REQUIRED or INSUFFICIENT_INFORMATION.

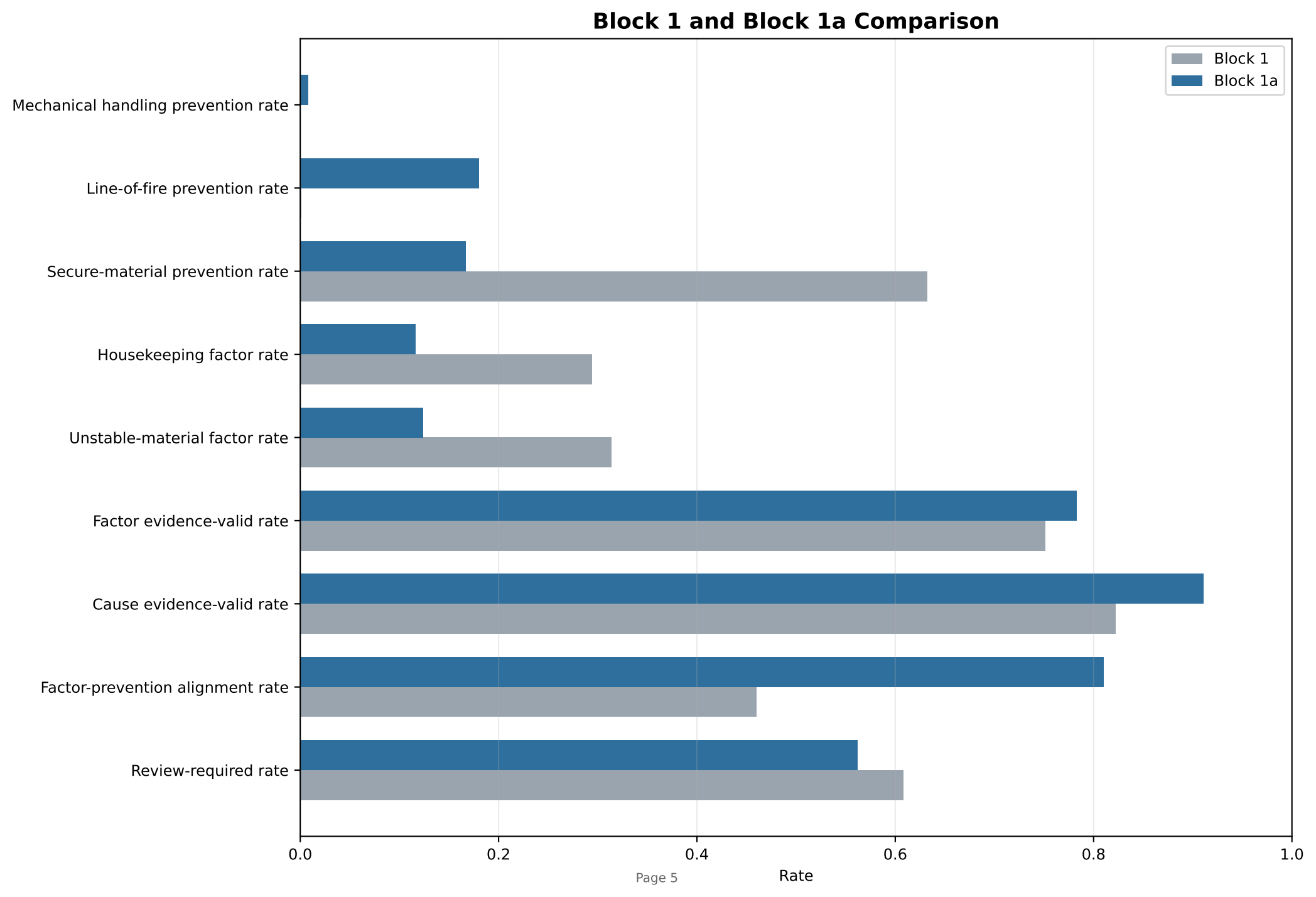
11. Added deterministic preferred factor-to-prevention mappings.

12. Added evidence checks that distinguish event evidence from factor
    evidence.

13. Improved routing of nonwork, natural, illness, noninjury, and
    insufficient-information records.

# Block 1 Versus Block 1a Metrics

| Measure | Block 1 | Block 1A | Change |
|---|---|---|---|
| Successful classifications | 0.9910 | 0.9450 | -0.0460 |
| Review-required rate | 0.6080 | 0.5620 | -0.0460 |
| Factor-prevention alignment rate | 0.4600 | 0.8100 | 0.3500 |
| Cause evidence-valid rate | 0.8220 | 0.9110 | 0.0890 |
| Factor evidence-valid rate | 0.7510 | 0.7830 | 0.0320 |
| No separate factor rate | 0.1090 | 0.2470 | 0.1380 |
| Unstable-material factor rate | 0.3140 | 0.1240 | -0.1900 |
| Housekeeping factor rate | 0.2940 | 0.1160 | -0.1780 |
| Secure-material prevention rate | 0.6320 | 0.1670 | -0.4650 |
| Line-of-fire prevention rate | 0.0010 | 0.1800 | 0.1790 |
| Mechanical handling prevention rate | 0.0000 | 0.0080 | 0.0080 |

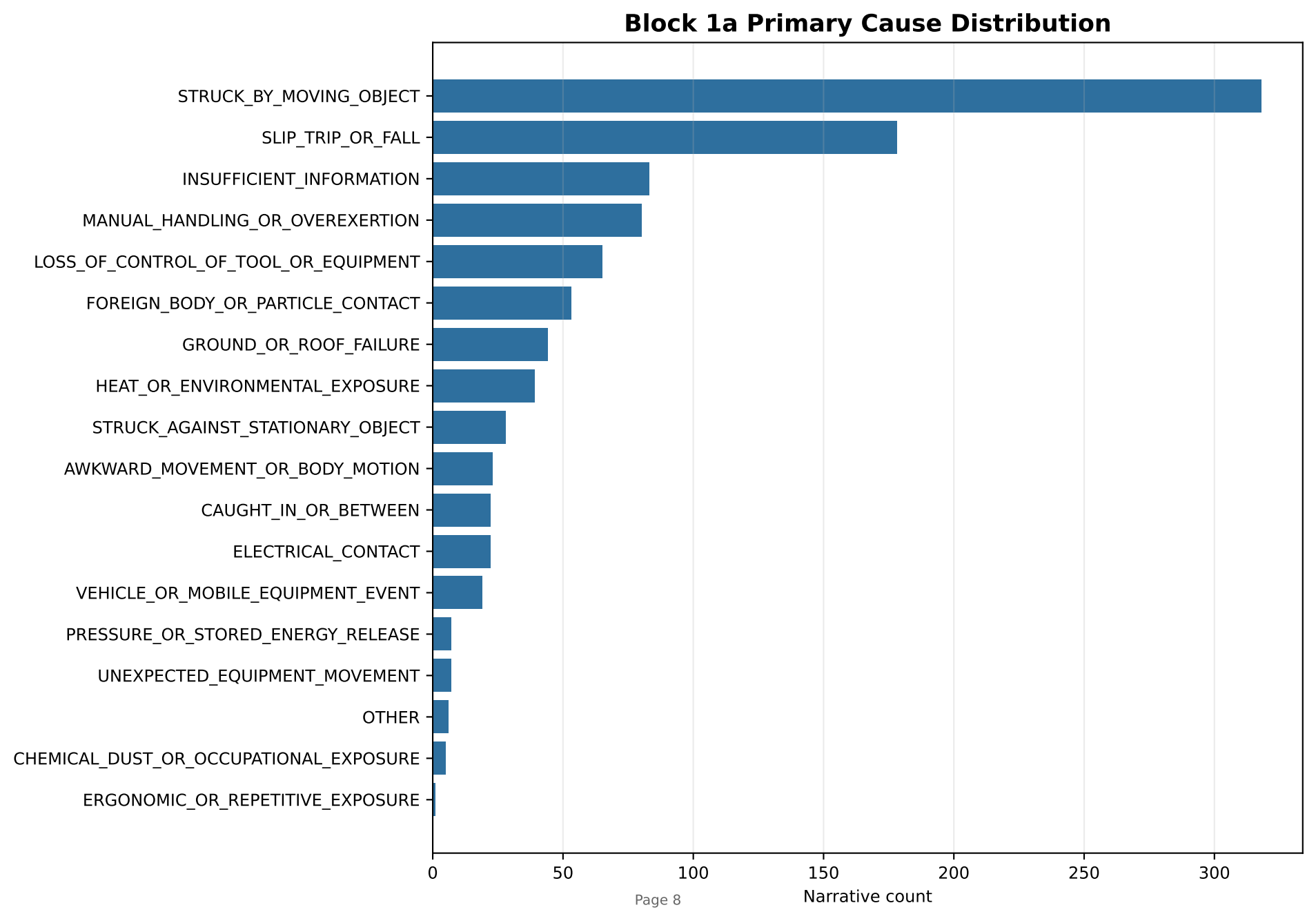

Block 1 and Block 1a Comparison
Block 1
Block 1a
Mechanical handling prevention rate
Line-of-fire prevention rate
Secure-material prevention rate
Housekeeping factor rate
Unstable-material factor rate
Factor evidence-valid rate
Cause evidence-valid rate
Factor-prevention alignment rate
Review-required rate
0.0
0.2
0.4
0.6
0.8
1.0
Rate

# Block 1a Record-Type Distribution

| Record Type | Count | Percent |
|---|---|---|
| ACUTE_ACCIDENT | 881 | 88.1000 |
| NONINJURY_REPORT | 62 | 6.2000 |
| OCCUPATIONAL_EXPOSURE_OR_ILLNESS | 25 | 2.5000 |
| NONWORK_RELATED_OR_NATURAL_EVENT | 19 | 1.9000 |
| INSUFFICIENT_EVENT_INFORMATION | 13 | 1.3000 |

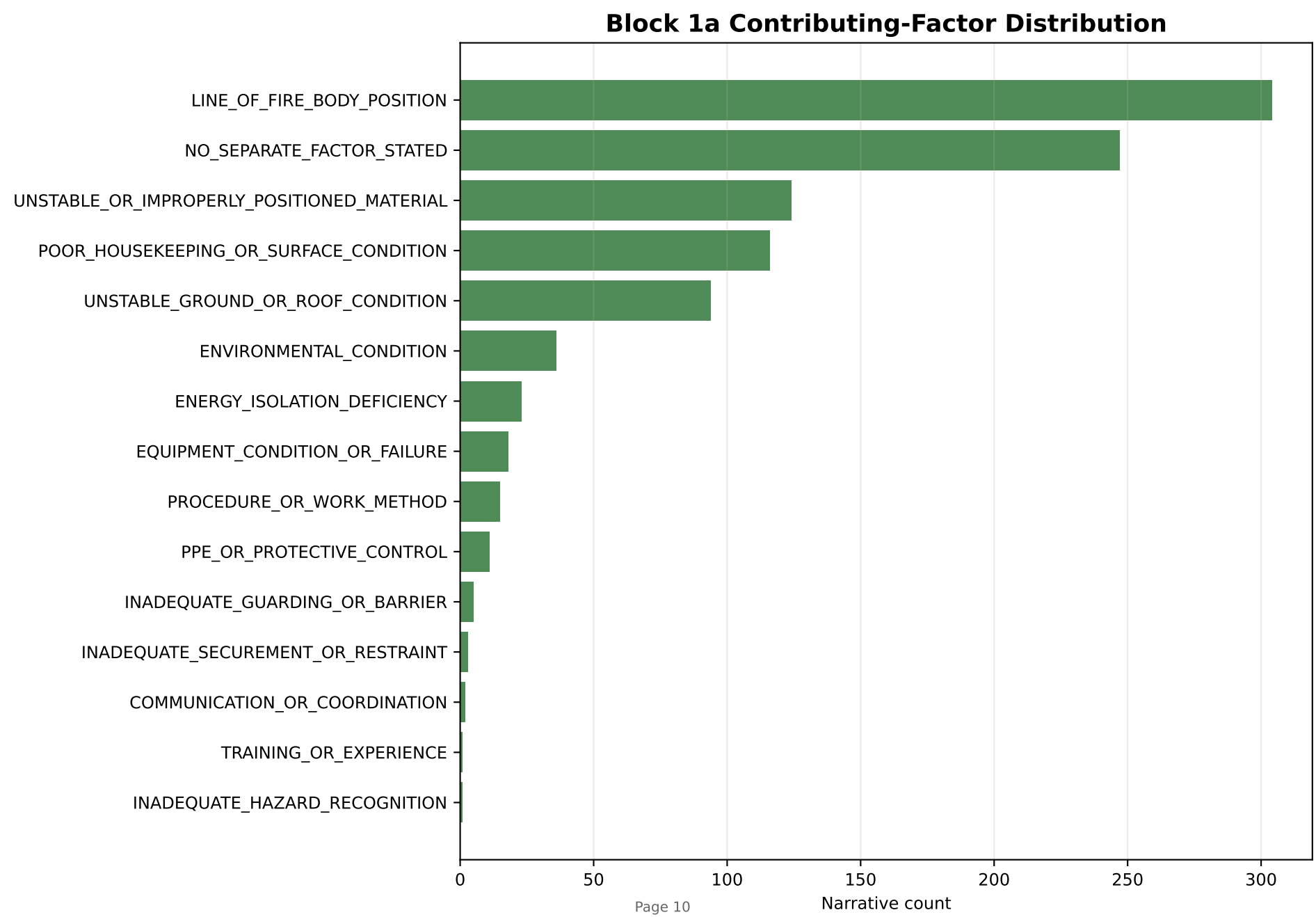
# Block 1a Primary Cause Distribution

| Primary Cause | Count | Percent |
|---|---|---|
| STRUCK_BY_MOVING_OBJECT | 318 | 31.8000 |
| SLIP_TRIP_OR_FALL | 178 | 17.8000 |
| INSUFFICIENT_INFORMATION | 83 | 8.3000 |
| MANUAL_HANDLING_OR_OVEREXERTION | 80 | 8.0000 |
| LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT | 65 | 6.5000 |
| FOREIGN_BODY_OR_PARTICLE_CONTACT | 53 | 5.3000 |
| GROUND_OR_ROOF_FAILURE | 44 | 4.4000 |
| HEAT_OR_ENVIRONMENTAL_EXPOSURE | 39 | 3.9000 |
| STRUCK_AGAINST_STATIONARY_OBJECT | 28 | 2.8000 |
| AWKWARD_MOVEMENT_OR_BODY_MOTION | 23 | 2.3000 |
| CAUGHT_IN_OR_BETWEEN | 22 | 2.2000 |
| ELECTRICAL_CONTACT | 22 | 2.2000 |
| VEHICLE_OR_MOBILE_EQUIPMENT_EVENT | 19 | 1.9000 |
| UNEXPECTED_EQUIPMENT_MOVEMENT | 7 | 0.7000 |
| PRESSURE_OR_STORED_ENERGY_RELEASE | 7 | 0.7000 |
| OTHER | 6 | 0.6000 |
| CHEMICAL_DUST_OR_OCCUPATIONAL_EXPOSURE | 5 | 0.5000 |
| ERGONOMIC_OR_REPETITIVE_EXPOSURE | 1 | 0.1000 |


Block 1a Primary Cause Distribution
STRUCK_BY_MOVING_OBJECT
SLIP_TRIP_OR_FALL
INSUFFICIENT_INFORMATION
MANUAL_HANDLING_OR_OVEREXERTION
LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
FOREIGN_BODY_OR_PARTICLE_CONTACT
GROUND_OR_ROOF_FAILURE
HEAT_OR_ENVIRONMENTAL_EXPOSURE
STRUCK_AGAINST_STATIONARY_OBJECT
AWKWARD_MOVEMENT_OR_BODY_MOTION
CAUGHT_IN_OR_BETWEEN
ELECTRICAL_CONTACT
VEHICLE_OR_MOBILE_EQUIPMENT_EVENT
PRESSURE_OR_STORED_ENERGY_RELEASE
UNEXPECTED_EQUIPMENT_MOVEMENT
OTHER
CHEMICAL_DUST_OR_OCCUPATIONAL_EXPOSURE
ERGONOMIC_OR_REPETITIVE_EXPOSURE
0
50
100
150
200
250
300
Narrative count

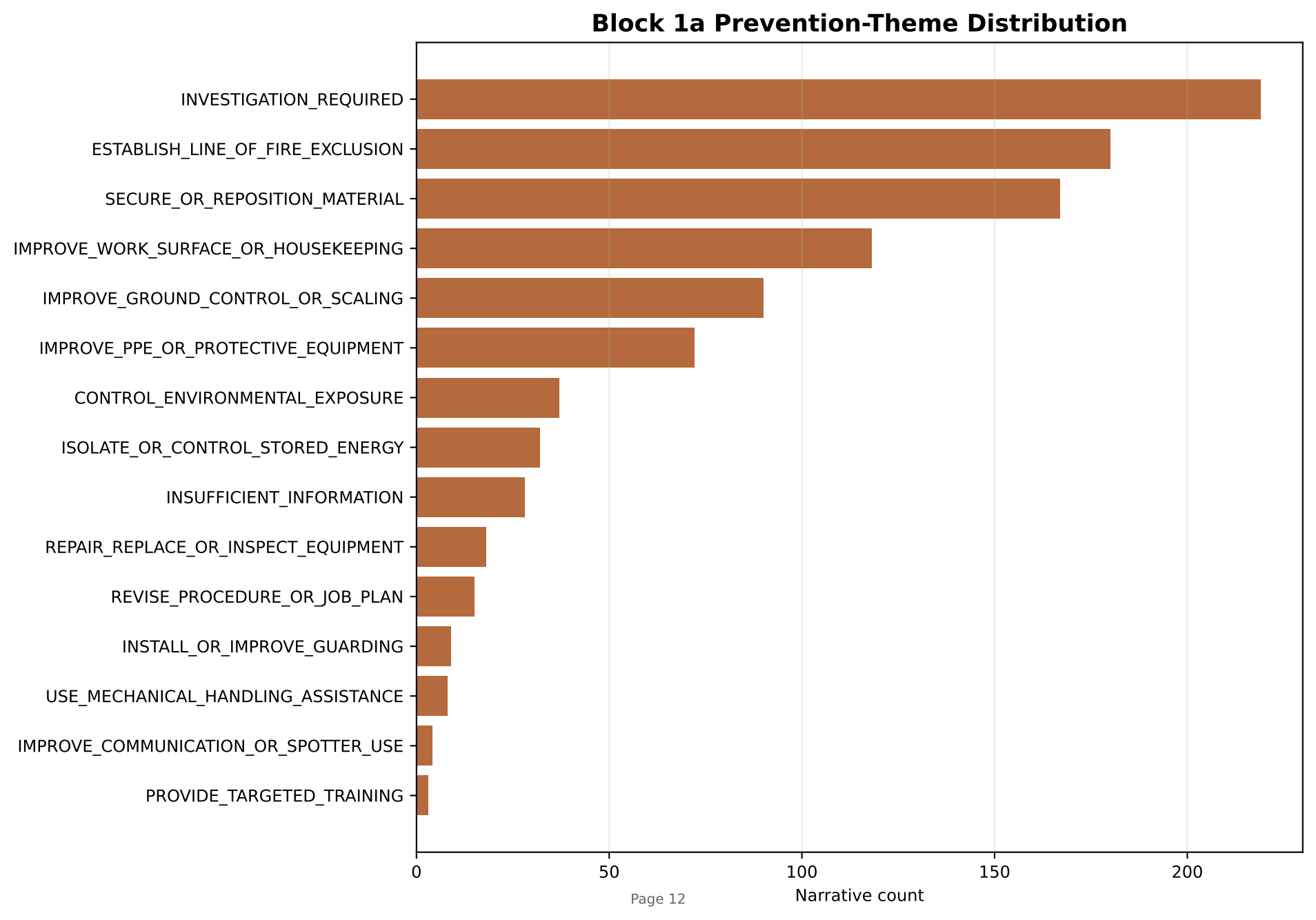
# Block 1a Contributing-Factor Distribution

| Primary Contributing Factor | Count | Percent |
|---|---|---|
| LINE_OF_FIRE_BODY_POSITION | 304 | 30.4000 |
| NO_SEPARATE_FACTOR_STATED | 247 | 24.7000 |
| UNSTABLE_OR_IMPROPERLY_POSITIONED_MATER | 124 | 12.4000 |
| POOR_HOUSEKEEPING_OR_SURFACE_CONDITION | 116 | 11.6000 |
| UNSTABLE_GROUND_OR_ROOF_CONDITION | 94 | 9.4000 |
| ENVIRONMENTAL_CONDITION | 36 | 3.6000 |
| ENERGY_ISOLATION_DEFICIENCY | 23 | 2.3000 |
| EQUIPMENT_CONDITION_OR_FAILURE | 18 | 1.8000 |
| PROCEDURE_OR_WORK_METHOD | 15 | 1.5000 |
| PPE_OR_PROTECTIVE_CONTROL | 11 | 1.1000 |
| INADEQUATE_GUARDING_OR_BARRIER | 5 | 0.5000 |
| INADEQUATE_SECUREMENT_OR_RESTRAINT | 3 | 0.3000 |
| COMMUNICATION_OR_COORDINATION | 2 | 0.2000 |
| INADEQUATE_HAZARD_RECOGNITION | 1 | 0.1000 |
| TRAINING_OR_EXPERIENCE | 1 | 0.1000 |


Block 1a Contributing-Factor Distribution
LINE_OF_FIRE_BODY_POSITION
NO_SEPARATE_FACTOR_STATED
UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
UNSTABLE_GROUND_OR_ROOF_CONDITION
ENVIRONMENTAL_CONDITION
ENERGY_ISOLATION_DEFICIENCY
EQUIPMENT_CONDITION_OR_FAILURE
PROCEDURE_OR_WORK_METHOD
PPE_OR_PROTECTIVE_CONTROL
INADEQUATE_GUARDING_OR_BARRIER
INADEQUATE_SECUREMENT_OR_RESTRAINT
COMMUNICATION_OR_COORDINATION
TRAINING_OR_EXPERIENCE
INADEQUATE_HAZARD_RECOGNITION
0
50
100
150
200
250
300
Narrative count

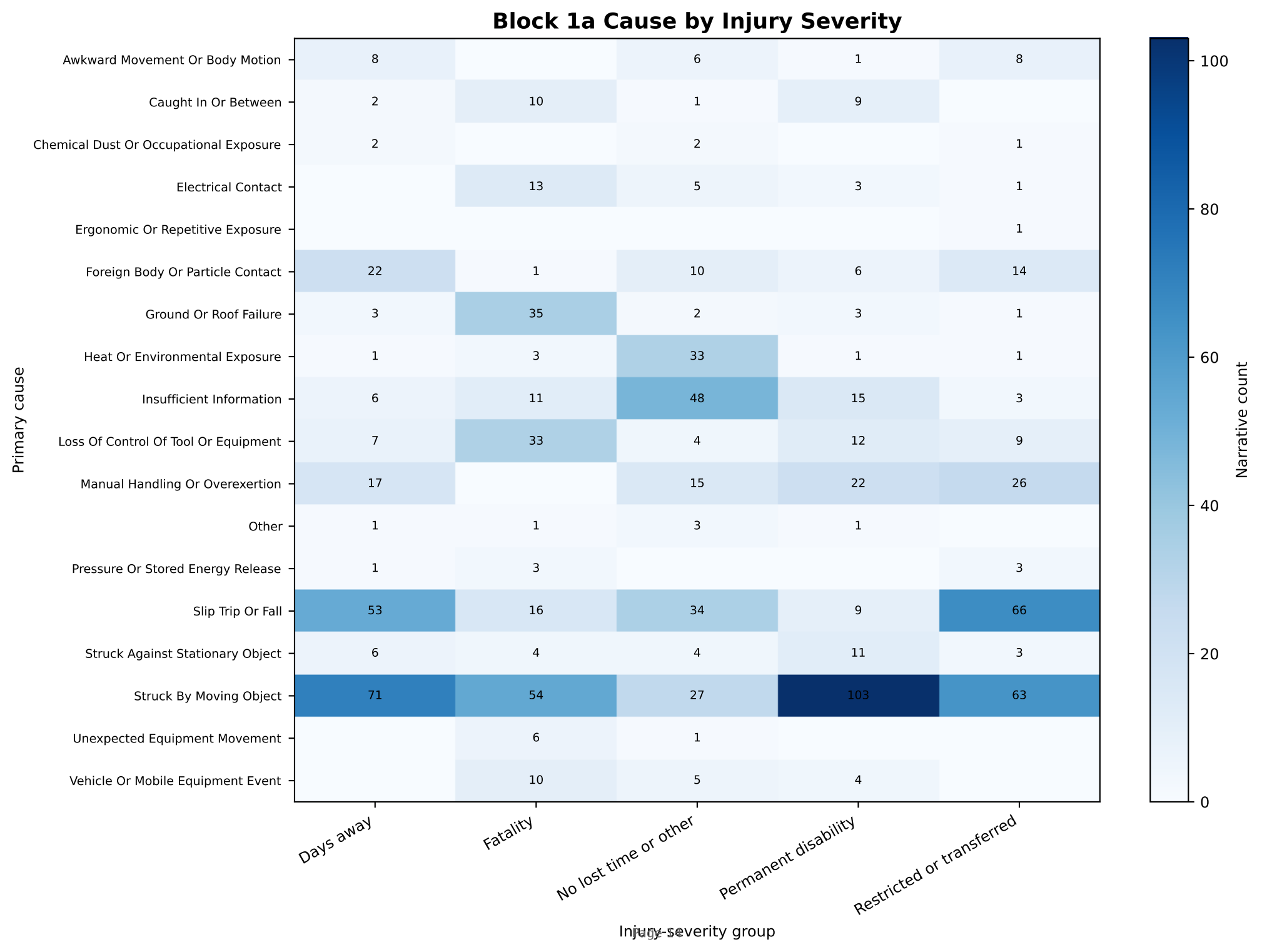
# Block 1a Prevention-Theme Distribution

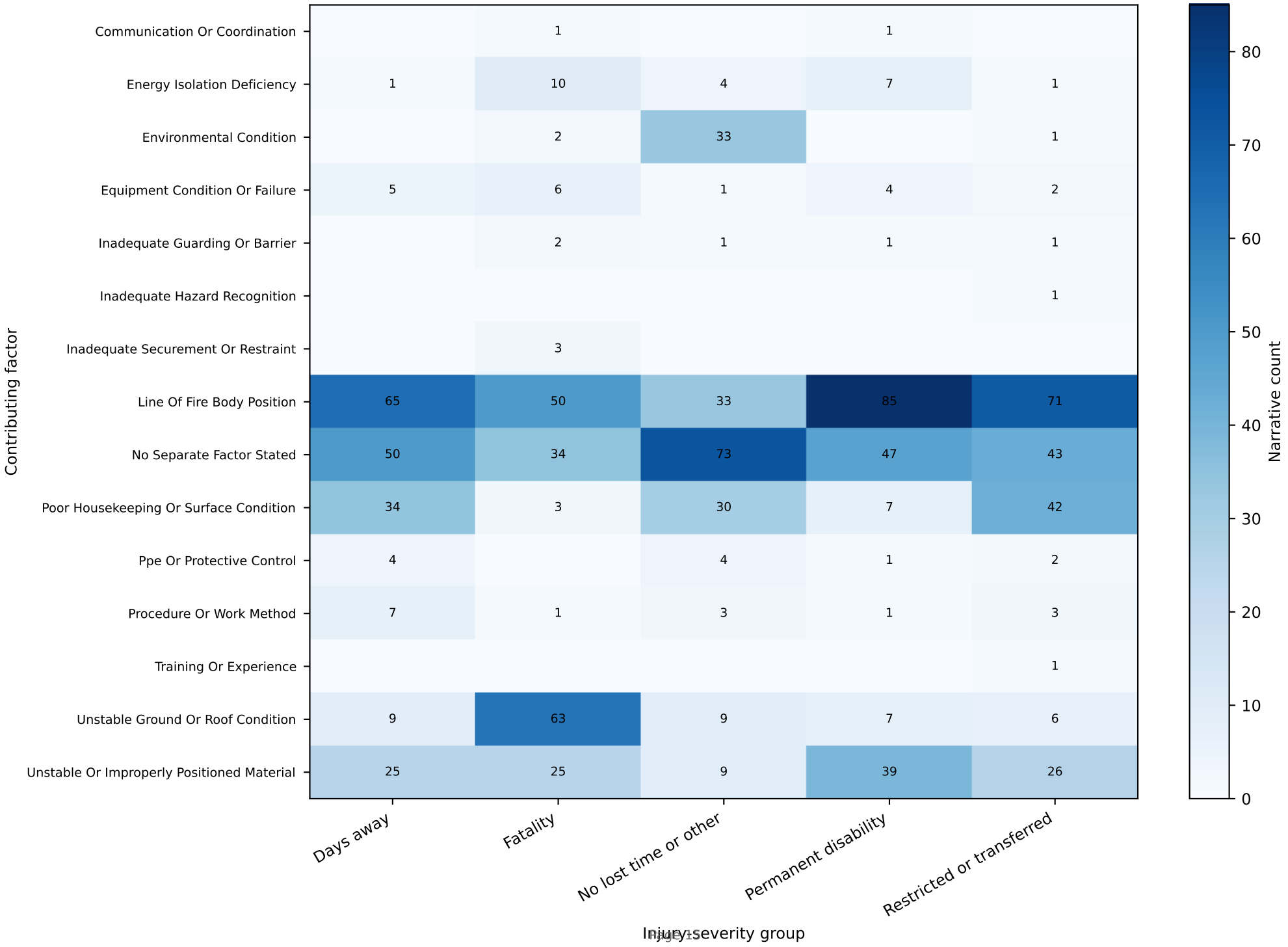
| Prevention Theme | Count | Percent |
|---|---|---|
| INVESTIGATION_REQUIRED | 219 | 21.9000 |
| ESTABLISH_LINE_OF_FIRE_EXCLUSION | 180 | 18.0000 |
| SECURE_OR_REPOSITION_MATERIAL | 167 | 16.7000 |
| IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING | 118 | 11.8000 |
| IMPROVE_GROUND_CONTROL_OR_SCALING | 90 | 9.0000 |
| IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT | 72 | 7.2000 |
| CONTROL_ENVIRONMENTAL_EXPOSURE | 37 | 3.7000 |
| ISOLATE_OR_CONTROL_STORED_ENERGY | 32 | 3.2000 |
| INSUFFICIENT_INFORMATION | 28 | 2.8000 |
| REPAIR_REPLACE_OR_INSPECT_EQUIPMENT | 18 | 1.8000 |
| REVISE_PROCEDURE_OR_JOB_PLAN | 15 | 1.5000 |
| INSTALL_OR_IMPROVE_GUARDING | 9 | 0.9000 |
| USE_MECHANICAL_HANDLING_ASSISTANCE | 8 | 0.8000 |
| IMPROVE_COMMUNICATION_OR_SPOTTER_USE | 4 | 0.4000 |
| PROVIDE_TARGETED_TRAINING | 3 | 0.3000 |

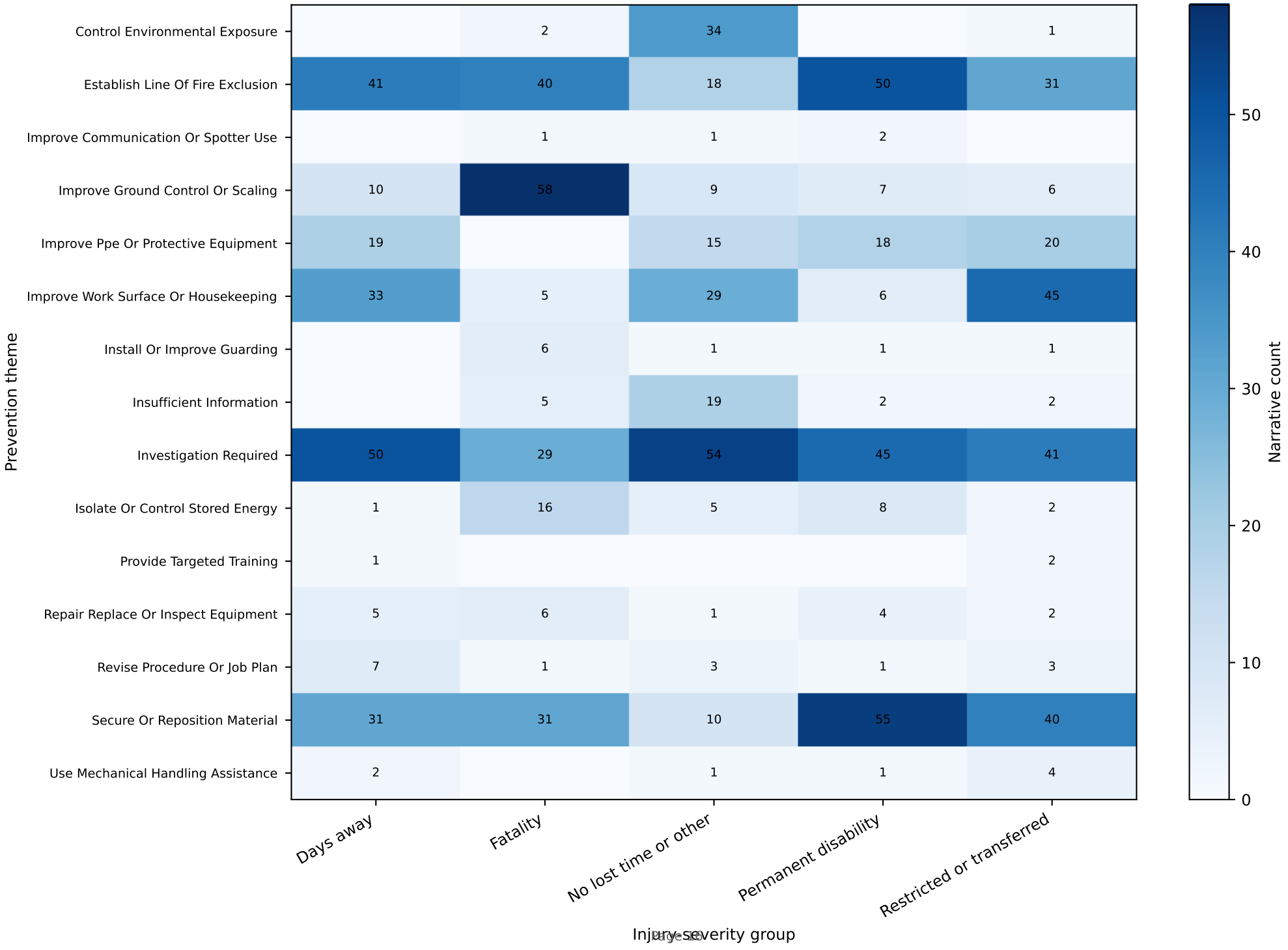

Block 1a Prevention-Theme Distribution
INVESTIGATION_REQUIRED
ESTABLISH_LINE_OF_FIRE_EXCLUSION
SECURE_OR_REPOSITION_MATERIAL
IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
IMPROVE_GROUND_CONTROL_OR_SCALING
IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
CONTROL_ENVIRONMENTAL_EXPOSURE
ISOLATE_OR_CONTROL_STORED_ENERGY
INSUFFICIENT_INFORMATION
REPAIR_REPLACE_OR_INSPECT_EQUIPMENT
REVISE_PROCEDURE_OR_JOB_PLAN
INSTALL_OR_IMPROVE_GUARDING
USE_MECHANICAL_HANDLING_ASSISTANCE
IMPROVE_COMMUNICATION_OR_SPOTTER_USE
PROVIDE_TARGETED_TRAINING
0
50
100
150
200
Narrative count

# Block 1a Automated Review Reasons

| Review Reason | Count |
|---|---|
| At least one support level is low | 216 |
| At least one confidence value is low | 197 |
| No separate contributing factor stated | 192 |
| Factor evidence not found in narrative | 162 |
| Factor-prevention combination is outside allowed map | 135 |
| Record is outside primary acute-accident population | 119 |
| Model response failed validation | 55 |
| Alternative prevention selected instead of preferred mapping | 31 |
| Cause is insufficient information | 28 |
| Cause evidence not found in narrative | 14 |
| Narrative may be truncated | 11 |


Block 1a Cause by Injury Severity
Primary cause
Awkward Movement Or Body Motion
Caught In Or Between
Chemical Dust Or Occupational Exposure
Electrical Contact
Ergonomic Or Repetitive Exposure
Foreign Body Or Particle Contact
Ground Or Roof Failure
Heat Or Environmental Exposure
Insufficient Information
Loss Of Control Of Tool Or Equipment
Manual Handling Or Overexertion
Other
Pressure Or Stored Energy Release
Slip Trip Or Fall
Struck Against Stationary Object
Struck By Moving Object
Unexpected Equipment Movement
Vehicle Or Mobile Equipment Event
8 6 1 8
2 10 1 9
2 2 1
13 5 3 1
1
22 1 10 6 14
3 35 2 3 1
1 3 33 1 1
6 11 48 15 3
7 33 4 12 9
17 15 22 26
1 1 3 1
1 3 3
53 16 34 9 66
6 4 4 11 3
71 54 27 103 63
6 1
10 5 4
Days away
Fatality
No lost time or other
Permanent disability
Restricted or transferred
Injury-severity group
Narrative count
100
80
60
40
20
0


Block 1a Factor by Injury Severity
Contributing factor
Communication Or Coordination
Energy Isolation Deficiency
Environmental Condition
Equipment Condition Or Failure
Inadequate Guarding Or Barrier
Inadequate Hazard Recognition
Inadequate Securement Or Restraint
Line Of Fire Body Position
No Separate Factor Stated
Poor Housekeeping Or Surface Condition
Ppe Or Protective Control
Procedure Or Work Method
Training Or Experience
Unstable Ground Or Roof Condition
Unstable Or Improperly Positioned Material
1 1
1 10 4 7 1
2 33 1
5 6 1 4 2
2 1 1 1
1
3
65 50 33 85 71
50 34 73 47 43
34 3 30 7 42
4 4 1 2
7 1 3 1 3
1
9 63 9 7 6
25 25 9 39 26
Days away
Fatality
No lost time or other
Permanent disability
Restricted or transferred
Injury severity group
Narrative count
0
10
20
30
40
50
60
70
80


Block 1a Prevention Theme by Injury Severity
Prevention theme
Control Environmental Exposure
Establish Line Of Fire Exclusion
Improve Communication Or Spotter Use
Improve Ground Control Or Scaling
Improve Ppe Or Protective Equipment
Improve Work Surface Or Housekeeping
Install Or Improve Guarding
Insufficient Information
Investigation Required
Isolate Or Control Stored Energy
Provide Targeted Training
Repair Replace Or Inspect Equipment
Revise Procedure Or Job Plan
Secure Or Reposition Material
Use Mechanical Handling Assistance
Days away
Fatality
No lost time or other
Permanent disability
Restricted or transferred
Injury severity group
Narrative count
0
10
20
30
40
50
2 34 1
41 40 18 50 31
1 1 2
10 58 9 7 6
19 15 18 20
33 5 29 6 45
6 1 1 1
5 19 2 2
50 29 54 45 41
1 16 5 8 2
1 2
5 6 1 4 2
7 1 3 1 3
31 31 10 55 40
2 1 1 4


Block 1a Factor and Prevention Relationships
Contributing factor
Communication Or Coordination
Energy Isolation Deficiency
Environmental Condition
Equipment Condition Or Failure
Inadequate Guarding Or Barrier
Inadequate Hazard Recognition
Inadequate Securement Or Restraint
Line Of Fire Body Position
No Separate Factor Stated
Poor Housekeeping Or Surface Condition
Ppe Or Protective Control
Procedure Or Work Method
Training Or Experience
Unstable Ground Or Roof Condition
Unstable Or Improperly Positioned Material
2
23
36
18
5
1
3
180
2
2
58
7
7
1
41
6
28
219
1
3
110
2
11
15
1
88
1
4
1
1
123
CONTROL_ENVIRONMENTAL_EXPOSURE
ESTABLISH_LINE_OF_FIRE_EXCLUSION
IMPROVE_COMMUNICATION_OR_SPOTTER_USE
IMPROVE_GROUND_CONTROL_OR_SCALING
IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
INSTALL_OR_IMPROVE_GUARDING
INSUFFICIENT_INFORMATION
INVESTIGATION_REQUIRED
ISOLATE_OR_CONTROL_STORED_ENERGY
PROVIDE_TARGETED_TRAINING
REPAIR_REPLACE_OR_INSPECT_EQUIPMENT
REVISE_PROCEDURE_OR_JOB_PLAN
SECURE_OR_REPOSITION_MATERIAL
USE_MECHANICAL_HANDLING_ASSISTANCE
Prevention theme
Narrative count
0
25
50
75
100
125
150
175
200

# Label and Consistency Changes

| Measure | Count |
|---|---|
| Cause labels changed | 546 |
| Factor labels changed | 612 |
| Prevention labels changed | 690 |
| Block 1 alignment failures corrected | 15 |
| Block 1a automatic consistency repairs | 156 |

# Examples with Revised Contributing Factors - 1 of 5

These examples show narratives for which the contributing-factor label changed between Block 1 and Block 1a. Review whether the new label is better supported by a separate factor-evidence excerpt.

==========
Example 1 | Record 220253350005 | Restricted or transferred
Block 1 cause: STRUCK_BY_MOVING_OBJECT
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_GROUND_CONTROL_OR_SCALING
Block 1a review reasons: Factor evidence not found in narrative

Narrative:
While standing at the control panel of a portable screen a large rock fell from the top surface and struck the top of EE's hard hat.  The hard hat shell contacted EE's head, causing a cut to EE's scalp. The impact knocked EE down and caused pain to EE's back and right shoulder.

Block 1a cause evidence: A large rock fell from the top surface and struck the top of EE's hard hat.
Block 1a factor evidence: The rock fell from the top surface, indicating an unstable condition.
Block 1a prevention rationale: Improving ground control would prevent rocks from falling and injuring workers.

==========
Example 2 | Record 220211340010 | Restricted or transferred
Block 1 cause: AWKWARD_MOVEMENT_OR_BODY_MOTION
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: LINE_OF_FIRE_BODY_POSITION
Block 1 prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Block 1a prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Block 1a review reasons: Factor-prevention combination is outside allowed map

Narrative:
Operator was traveling down a roadway in a haul truck when struck a pot hole. Operator felt discomfort in back and the incident cause lower back pain and spasms. Incident did not become restricted or reportable till 5/4/2021 after further evaluation from healthcare provider.

Block 1a cause evidence: Operator was traveling down a roadway in a haul truck when struck a pot hole.
Block 1a factor evidence: Operator felt discomfort in back and the incident caused lower back pain and spasms.
Block 1a prevention rationale: Improving the road surface to reduce potholes would prevent the truck from striking them.

==========
Example 3 | Record 220001640065 | Restricted or transferred
Block 1 cause: STRUCK_BY_MOVING_OBJECT
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION

# Examples with Revised Contributing Factors - 2 of 5

Block 1a factor: LINE_OF_FIRE_BODY_POSITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
Block 1a review reasons: Factor-prevention combination is outside allowed map

Narrative:
EMPLOYEE WAS APPLYING STOPPING SEALANT WITH A SPRAY MACHINE WHEN APPLICATION HOSE SLIPPED THROUGH HIS HANDS AND SEALANT STRUCK HIS FACE, NOSE, LEFT EYE AND EAR.

Block 1a cause evidence: EMPLOYEE WAS APPLYING STOPPING SEALANT WITH A SPRAY MACHINE WHEN APPLICATION HOSE SLIPPED THROUGH HIS HANDS AND SEALANT STRUCK HIS FACE, NOSE, LEFT EYE AND EAR.
Block 1a factor evidence: APPLICATION HOSE SLIPPED THROUGH HIS HANDS
Block 1a prevention rationale: Improving the securement of the application hose can prevent it from slipping and striking the employee.

==========
Example 4 | Record 220021710013 | Days away
Block 1 cause: OTHER
Block 1a cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: PROVIDE_TARGETED_TRAINING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
EE STATES HE WAS MOVING SPOOL OF CABLE IN SHOP AND FELT PAIN IN LOWER ABDOMEN. LATER DIAGNOSED AS HERNIA. HAS NOT MISSED ANY DAYS YET BUT IS ON RESTRICTED DUTY. HISTORY OF PREVIOUS HERNIAS.

Block 1a cause evidence: EE states he was moving spool of cable in shop and felt pain in lower abdomen.
Block 1a factor evidence:
Block 1a prevention rationale: Use mechanical assistance to reduce manual handling and overexertion.

==========
Example 5 | Record 220122060025 | Restricted or transferred
Block 1 cause: AWKWARD_MOVEMENT_OR_BODY_MOTION
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: PROCEDURE_OR_WORK_METHOD
Block 1a factor: LINE_OF_FIRE_BODY_POSITION
Block 1 prevention: REVISE_PROCEDURE_OR_JOB_PLAN
Block 1a prevention: USE_MECHANICAL_HANDLING_ASSISTANCE
Block 1a review reasons: Factor-prevention combination is outside allowed map

Narrative:
Worker was carrying a 35lb stone, when he twisted to put it on the stack he felt his left hip "pop". Left hip strain. Restricted duty, No lost time.

Block 1a cause evidence: Worker was carrying a 35lb stone, when he twisted to put it on the stack he

# Examples with Revised Contributing Factors - 3 of 5

felt his left hip 'pop'.
Block 1a factor evidence: Worker was carrying a 35lb stone, when he twisted to put it on the stack he felt his left hip 'pop'.
Block 1a prevention rationale: Using mechanical assistance to lift heavy objects can reduce the risk of hip strains.

==========
Example 6 | Record 220090090045 | Restricted or transferred
Block 1 cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1a cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1 factor: LINE_OF_FIRE_BODY_POSITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: PROVIDE_TARGETED_TRAINING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Employee was carrying a battery from the back of the truck to the track mobile. As he walked accross the railroad tracks, he felt pain on left side of his back.

Block 1a cause evidence: Employee was carrying a battery from the back of the truck to the track mobile.
Block 1a factor evidence:
Block 1a prevention rationale: Using mechanical assistance could reduce the risk of back strain during manual handling.

==========
Example 7 | Record 220122750008 | Days away
Block 1 cause: OTHER
Block 1a cause: FOREIGN_BODY_OR_PARTICLE_CONTACT
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Miner was shoveling dirt when a foreign object became lodged in his eye. Ophthalmologist removed the foreign object using a slit lamp. Antibiotic eye drops were prescribed.

Block 1a cause evidence: Miner was shoveling dirt when a foreign object became lodged in his eye.
Block 1a factor evidence:
Block 1a prevention rationale: Improving eye protection can prevent foreign objects from entering the eye.

==========
Example 8 | Record 220210560004 | Fatality
Block 1 cause: STRUCK_BY_MOVING_OBJECT

# Examples with Revised Contributing Factors - 4 of 5

Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Block 1a factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_GROUND_CONTROL_OR_SCALING
Block 1a review reasons: Factor evidence not found in narrative

Narrative:
A rail haulage offsider was riding back to the chute to get another load when the locomotive over traveled into the chute causing fatal injuries to the offsider rail employee.

Block 1a cause evidence: the locomotive over traveled into the chute
Block 1a factor evidence: the chute is an unstable ground condition
Block 1a prevention rationale: Improving the examination and support of the chute would prevent the locomotive from over traveling.

==========
Example 9 | Record 220113480014 | Fatality
Block 1 cause: UNEXPECTED_EQUIPMENT_MOVEMENT
Block 1a cause: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
Block 1 factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Block 1a factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_GROUND_CONTROL_OR_SCALING
Block 1a review reasons:

Narrative:
EE was shoving top soil away from the highwall. He backed up over and through an 8 foot berm that had been constructed at the edge of the highwall.  The dozer went over the wall rolling over and throwing him out of the cab.

Block 1a cause evidence: EE backed up over and through an 8 foot berm that had been constructed at the edge of the highwall.
Block 1a factor evidence: The dozer went over the wall rolling over and throwing him out of the cab.
Block 1a prevention rationale: Improving the examination and support of the berm would prevent the dozer from rolling over it.

==========
Example 10 | Record 220023600016 | Restricted or transferred
Block 1 cause: SLIP_TRIP_OR_FALL
Block 1a cause: SLIP_TRIP_OR_FALL
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:

# Examples with Revised Contributing Factors - 5 of 5

WHILE INSTALLING A FK PUMP SLIPPED AND STRUCK MINER ON RIGHT FOOT. AFTER 2 EXAMS A MRI WAS DONE AND DETERMINED THAT MINER HAD A HAIRLINE FRACTURE ON 12-19-02. THIS BECAME REPORTABLE ON DECEMBE R 19, 2002.

Block 1a cause evidence: WHILE INSTALLING A FK PUMP SLIPPED AND STRUCK MINER ON RIGHT FOOT.
Block 1a factor evidence:
Block 1a prevention rationale: Improving material stability and securing the pump would have prevented the slip.

==========
Example 11 | Record 220232500006 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: ENVIRONMENTAL_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
EE was working at the 24S welding and then went to help co-workers when EE was done. Upon return to the Shop employee began to feel nauseous and began vomiting. EE was taken to the ER to be seen. EE was treated for heat illness. EE was given a full release at the ER.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

==========
Example 12 | Record 220250210029 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: OTHER
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; Cause evidence not found in narrative; At least one confidence value is low; At least one support level is low

Narrative:
EE came to the shop and said side was hurting.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: Further investigation is needed to determine the cause and contributing factors.

# Examples with Revised Prevention Themes - 1 of 5

These examples show changes in prevention-theme selection after introducing the preferred factor-to-prevention mapping.

==========

Example 1 | Record 220253350005 | Restricted or transferred
Block 1 cause: STRUCK_BY_MOVING_OBJECT
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_GROUND_CONTROL_OR_SCALING
Block 1a review reasons: Factor evidence not found in narrative

Narrative:
While standing at the control panel of a portable screen a large rock fell from the top surface and struck the top of EE's hard hat.  The hard hat shell contacted EE's head, causing a cut to EE's scalp. The impact knocked EE down and caused pain to EE's back and right shoulder.

Block 1a cause evidence: A large rock fell from the top surface and struck the top of EE's hard hat.
Block 1a factor evidence: The rock fell from the top surface, indicating an unstable condition.
Block 1a prevention rationale: Improving ground control would prevent rocks from falling and injuring workers.

==========

Example 2 | Record 220001640065 | Restricted or transferred
Block 1 cause: STRUCK_BY_MOVING_OBJECT
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: LINE_OF_FIRE_BODY_POSITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
Block 1a review reasons: Factor-prevention combination is outside allowed map

Narrative:
EMPLOYEE WAS APPLYING STOPPING SEALANT WITH A SPRAY MACHINE WHEN APPLICATION HOSE SLIPPED THROUGH HIS HANDS AND SEALANT STRUCK HIS FACE, NOSE, LEFT EYE AND EAR.

Block 1a cause evidence: EMPLOYEE WAS APPLYING STOPPING SEALANT WITH A SPRAY MACHINE WHEN APPLICATION HOSE SLIPPED THROUGH HIS HANDS AND SEALANT STRUCK HIS FACE, NOSE, LEFT EYE AND EAR.
Block 1a factor evidence: APPLICATION HOSE SLIPPED THROUGH HIS HANDS
Block 1a prevention rationale: Improving the securement of the application hose can prevent it from slipping and striking the employee.

==========

Example 3 | Record 220021710013 | Days away
Block 1 cause: OTHER
Block 1a cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED

# Examples with Revised Prevention Themes - 2 of 5

Block 1 prevention: PROVIDE_TARGETED_TRAINING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
EE STATES HE WAS MOVING SPOOL OF CABLE IN SHOP AND FELT PAIN IN LOWER ABDOMEN. LATER DIAGNOSED AS HERNIA. HAS NOT MISSED ANY DAYS YET BUT IS ON RESTRICTED DUTY. HISTORY OF PREVIOUS HERNIAS.

Block 1a cause evidence: EE states he was moving spool of cable in shop and felt pain in lower abdomen.
Block 1a factor evidence:
Block 1a prevention rationale: Use mechanical assistance to reduce manual handling and overexertion.

==========
Example 4 | Record 220122060025 | Restricted or transferred
Block 1 cause: AWKWARD_MOVEMENT_OR_BODY_MOTION
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: PROCEDURE_OR_WORK_METHOD
Block 1a factor: LINE_OF_FIRE_BODY_POSITION
Block 1 prevention: REVISE_PROCEDURE_OR_JOB_PLAN
Block 1a prevention: USE_MECHANICAL_HANDLING_ASSISTANCE
Block 1a review reasons: Factor-prevention combination is outside allowed map

Narrative:
Worker was carrying a 35lb stone, when he twisted to put it on the stack he felt his left hip "pop". Left hip strain. Restricted duty, No lost time.

Block 1a cause evidence: Worker was carrying a 35lb stone, when he twisted to put it on the stack he felt his left hip 'pop'.
Block 1a factor evidence: Worker was carrying a 35lb stone, when he twisted to put it on the stack he felt his left hip 'pop'.
Block 1a prevention rationale: Using mechanical assistance to lift heavy objects can reduce the risk of hip strains.

==========
Example 5 | Record 220090090045 | Restricted or transferred
Block 1 cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1a cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1 factor: LINE_OF_FIRE_BODY_POSITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: PROVIDE_TARGETED_TRAINING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Employee was carrying a battery from the back of the truck to the track mobile. As he walked accross the railroad tracks, he felt pain on left side of his back.

# Examples with Revised Prevention Themes - 3 of 5

Block 1a cause evidence: Employee was carrying a battery from the back of the truck to the track mobile.
Block 1a factor evidence:
Block 1a prevention rationale: Using mechanical assistance could reduce the risk of back strain during manual handling.

==========
Example 6 | Record 220122750008 | Days away
Block 1 cause: OTHER
Block 1a cause: FOREIGN_BODY_OR_PARTICLE_CONTACT
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Miner was shoveling dirt when a foreign object became lodged in his eye. Ophthalmologist removed the foreign object using a slit lamp. Antibiotic eye drops were prescribed.

Block 1a cause evidence: Miner was shoveling dirt when a foreign object became lodged in his eye.
Block 1a factor evidence:
Block 1a prevention rationale: Improving eye protection can prevent foreign objects from entering the eye.

==========
Example 7 | Record 220210560004 | Fatality
Block 1 cause: STRUCK_BY_MOVING_OBJECT
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Block 1a factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_GROUND_CONTROL_OR_SCALING
Block 1a review reasons: Factor evidence not found in narrative

Narrative:
A rail haulage offsider was riding back to the chute to get another load when the locomotive over traveled into the chute causing fatal injuries to the offsider rail employee.

Block 1a cause evidence: the locomotive over traveled into the chute
Block 1a factor evidence: the chute is an unstable ground condition
Block 1a prevention rationale: Improving the examination and support of the chute would prevent the locomotive from over traveling.

==========
Example 8 | Record 220032580086 | No lost time or other
Block 1 cause: AWKWARD_MOVEMENT_OR_BODY_MOTION
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: LINE_OF_FIRE_BODY_POSITION

# Examples with Revised Prevention Themes - 4 of 5

Block 1a factor: LINE_OF_FIRE_BODY_POSITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
Block 1a review reasons: Factor-prevention combination is outside allowed map

Narrative:
EMPLOYEE WAS PACKING TYPE S ON PORTLAND WHEN TWO BAGS FELL OFF PACKING MACHINE ONTO THE CONVEYOR BELT BELOW. EMPLOYEE TRIED TO HOLD BACK ONE OF THE BAGS WITH HIS RIGHT FOOT TO PREVENT THE TWO BAGS FROM PLUGGING THE TRANSFER POINT. WHILE STRETCHING DOWN AND EXTENDING LEG TO HOLD BACK THE BAGS, HE FELT A POP IN LOWER BACK.

Block 1a cause evidence: EMPLOYEE WAS PACKING TYPE S ON PORTLAND WHEN TWO BAGS FELL OFF PACKING MACHINE ONTO THE CONVEYOR BELT BELOW.
Block 1a factor evidence: WHILE STRETCHING DOWN AND EXTENDING LEG TO HOLD BACK THE BAGS, HE FELT A POP IN LOWER BACK.
Block 1a prevention rationale: Improving protective equipment can prevent employees from placing their body in the line of fire of falling objects.

==========
Example 9 | Record 220113480014 | Fatality
Block 1 cause: UNEXPECTED_EQUIPMENT_MOVEMENT
Block 1a cause: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
Block 1 factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Block 1a factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_GROUND_CONTROL_OR_SCALING
Block 1a review reasons:

Narrative:
EE was shoving top soil away from the highwall. He backed up over and through an 8 foot berm that had been constructed at the edge of the highwall.  The dozer went over the wall rolling over and throwing him out of the cab.

Block 1a cause evidence: EE backed up over and through an 8 foot berm that had been constructed at the edge of the highwall.
Block 1a factor evidence: The dozer went over the wall rolling over and throwing him out of the cab.
Block 1a prevention rationale: Improving the examination and support of the berm would prevent the dozer from rolling over it.

==========
Example 10 | Record 220023600016 | Restricted or transferred
Block 1 cause: SLIP_TRIP_OR_FALL
Block 1a cause: SLIP_TRIP_OR_FALL
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

# Examples with Revised Prevention Themes - 5 of 5

Narrative:
WHILE INSTALLING A FK PUMP SLIPPED AND STRUCK MINER ON RIGHT FOOT. AFTER 2 EXAMS A MRI WAS DONE AND DETERMINED THAT MINER HAD A HAIRLINE FRACTURE ON 12-19-02. THIS BECAME REPORTABLE ON DECEMBE R 19, 2002.

Block 1a cause evidence: WHILE INSTALLING A FK PUMP SLIPPED AND STRUCK MINER ON RIGHT FOOT.
Block 1a factor evidence:
Block 1a prevention rationale: Improving material stability and securing the pump would have prevented the slip.

==========
Example 11 | Record 220232500006 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: ENVIRONMENTAL_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
EE was working at the 24S welding and then went to help co-workers when EE was done. Upon return to the Shop employee began to feel nauseous and began vomiting. EE was taken to the ER to be seen. EE was treated for heat illness. EE was given a full release at the ER.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

==========
Example 12 | Record 220250210029 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: OTHER
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; Cause evidence not found in narrative; At least one confidence value is low; At least one support level is low

Narrative:
EE came to the shop and said side was hurting.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: Further investigation is needed to determine the cause and contributing factors.

# Corrected Factor-Prevention Alignment Examples - 1 of 6

These records were outside the Block 1 compatibility map but are aligned under Block 1a. Confirm that the change reflects an improved factor or prevention decision rather than a mechanical remapping.

==========

Example 1 | Record 220253350005 | Restricted or transferred
Block 1 cause: STRUCK_BY_MOVING_OBJECT
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_GROUND_CONTROL_OR_SCALING
Block 1a review reasons: Factor evidence not found in narrative

Narrative:
While standing at the control panel of a portable screen a large rock fell from the top surface and struck the top of EE's hard hat.  The hard hat shell contacted EE's head, causing a cut to EE's scalp. The impact knocked EE down and caused pain to EE's back and right shoulder.

Block 1a cause evidence: A large rock fell from the top surface and struck the top of EE's hard hat.
Block 1a factor evidence: The rock fell from the top surface, indicating an unstable condition.
Block 1a prevention rationale: Improving ground control would prevent rocks from falling and injuring workers.

==========

Example 2 | Record 220021710013 | Days away
Block 1 cause: OTHER
Block 1a cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: PROVIDE_TARGETED_TRAINING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
EE STATES HE WAS MOVING SPOOL OF CABLE IN SHOP AND FELT PAIN IN LOWER ABDOMEN. LATER DIAGNOSED AS HERNIA. HAS NOT MISSED ANY DAYS YET BUT IS ON RESTRICTED DUTY. HISTORY OF PREVIOUS HERNIAS.

Block 1a cause evidence: EE states he was moving spool of cable in shop and felt pain in lower abdomen.
Block 1a factor evidence:
Block 1a prevention rationale: Use mechanical assistance to reduce manual handling and overexertion.

==========

Example 3 | Record 220090090045 | Restricted or transferred
Block 1 cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1a cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1 factor: LINE_OF_FIRE_BODY_POSITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: PROVIDE_TARGETED_TRAINING

# Corrected Factor-Prevention Alignment Examples - 2 of 6

Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Employee was carrying a battery from the back of the truck to the track mobile. As he walked accross the railroad tracks, he felt pain on left side of his back.

Block 1a cause evidence: Employee was carrying a battery from the back of the truck to the track mobile.
Block 1a factor evidence:
Block 1a prevention rationale: Using mechanical assistance could reduce the risk of back strain during manual handling.

==========
Example 4 | Record 220122750008 | Days away
Block 1 cause: OTHER
Block 1a cause: FOREIGN_BODY_OR_PARTICLE_CONTACT
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: IMPROVE_PPE_OR_PROTECTIVE_EQUIPMENT
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Miner was shoveling dirt when a foreign object became lodged in his eye. Ophthalmologist removed the foreign object using a slit lamp. Antibiotic eye drops were prescribed.

Block 1a cause evidence: Miner was shoveling dirt when a foreign object became lodged in his eye.
Block 1a factor evidence:
Block 1a prevention rationale: Improving eye protection can prevent foreign objects from entering the eye.

==========
Example 5 | Record 220023600016 | Restricted or transferred
Block 1 cause: SLIP_TRIP_OR_FALL
Block 1a cause: SLIP_TRIP_OR_FALL
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
WHILE INSTALLING A FK PUMP SLIPPED AND STRUCK MINER ON RIGHT FOOT. AFTER 2 EXAMS A MRI WAS DONE AND DETERMINED THAT MINER HAD A HAIRLINE FRACTURE ON 12-19-02. THIS BECAME REPORTABLE ON DECEMBE R 19, 2002.

Block 1a cause evidence: WHILE INSTALLING A FK PUMP SLIPPED AND STRUCK MINER ON RIGHT FOOT.

# Corrected Factor-Prevention Alignment Examples - 3 of 6

Block 1a factor evidence:
Block 1a prevention rationale: Improving material stability and securing the pump would have prevented the slip.

==========
Example 6 | Record 220250210029 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: OTHER
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; Cause evidence not found in narrative; At least one confidence value is low; At least one support level is low

Narrative:
EE came to the shop and said side was hurting.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: Further investigation is needed to determine the cause and contributing factors.

==========
Example 7 | Record 220162160052 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: HEAT_OR_ENVIRONMENTAL_EXPOSURE
Block 1 factor: NOT_STATED
Block 1a factor: ENVIRONMENTAL_CONDITION
Block 1 prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW
Block 1a prevention: CONTROL_ENVIRONMENTAL_EXPOSURE
Block 1a review reasons: Factor evidence not found in narrative; At least one confidence value is low; At least one support level is low

Narrative:
EE was working outside using a 20lb sledge hammer to loosen material that was stuck inside the chute. EE went in to cool off. About 10 minutes after EE tried calling to get ahold of someone because EE stated was cramping up and skin was hot clammy and EE was shivering.

Block 1a cause evidence: EE stated was cramping up and skin was hot clammy and EE was shivering.
Block 1a factor evidence:
Block 1a prevention rationale: Proper hydration and rest breaks are needed to prevent heat-related illnesses.

==========
Example 8 | Record 220192830020 | Restricted or transferred
Block 1 cause: AWKWARD_MOVEMENT_OR_BODY_MOTION
Block 1a cause: SLIP_TRIP_OR_FALL

# Corrected Factor-Prevention Alignment Examples - 4 of 6

Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Block 1a review reasons:

Narrative:
EE was carrying a piece of roof screen beside the roof bolter. EE lost footing and twisted knee.

Block 1a cause evidence: EE lost footing and twisted knee.
Block 1a factor evidence: EE was carrying a piece of roof screen beside the roof bolter.
Block 1a prevention rationale: Improving the walking surface and housekeeping can prevent slips and falls.

==========
Example 9 | Record 220262470030 | Permanent disability
Block 1 cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1a cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Lifting, pulling, pushing equipment (hoses and pumps) to and from a blasting area to prevent damage. Which caused a hernia for the miner.

Block 1a cause evidence: Lifting, pulling, pushing equipment (hoses and pumps) to and from a blasting area to prevent damage.
Block 1a factor evidence:
Block 1a prevention rationale: Use mechanical assistance to reduce manual handling and overexertion.

==========
Example 10 | Record 220021750028 | Permanent disability
Block 1 cause: STRUCK_BY_MOVING_OBJECT
Block 1a cause: STRUCK_BY_MOVING_OBJECT
Block 1 factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Block 1a factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: IMPROVE_GROUND_CONTROL_OR_SCALING
Block 1a review reasons:

Narrative:
EE WAS ATTEMPTING TO SPOT A ROOF BOLT ON THE CORNER OF A INTERSECTION IN #6 ENTRY DUE TO THE SLOPE OF ROCK HE PLACED HIS HAND ON DRILL STEEL TO START HOLE. ROCK FELL OUT STRIKING LEFT HAND.

Block 1a cause evidence: ROCK FELL OUT STRIKING LEFT HAND.

# Corrected Factor-Prevention Alignment Examples - 5 of 6

Block 1a factor evidence: HE PLACED HIS HAND ON DRILL STEEL TO START HOLE. ROCK FELL OUT STRIKING LEFT HAND.
Block 1a prevention rationale: Improving ground control and scaling would help prevent rock falls in steep areas.

================================================================================
Example 11 | Record 220080040010 | Permanent disability
Block 1 cause: AWKWARD_MOVEMENT_OR_BODY_MOTION
Block 1a cause: MANUAL_HANDLING_OR_OVEREXERTION
Block 1 factor: LINE_OF_FIRE_BODY_POSITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: PROVIDE_TARGETED_TRAINING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Record is outside primary acute-accident population; No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Employee was helping other employees maneuver outby Pinner cable and plug over other cables near the section power center.  As he was pulling the plug over another cable he felt a sharp pain in his abdomen.  Later diagnosed as a hernia.

Block 1a cause evidence: As he was pulling the plug over another cable he felt a sharp pain in his abdomen.
Block 1a factor evidence:
Block 1a prevention rationale: Using mechanical assistance would have reduced the manual handling and overexertion exposure.

================================================================================
Example 12 | Record 220152720007 | No lost time or other
Block 1 cause: PRESSURE_OR_STORED_ENERGY_RELEASE
Block 1a cause: ELECTRICAL_CONTACT
Block 1 factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Block 1a factor: ENERGY_ISOLATION_DEFICIENCY
Block 1 prevention: ISOLATE_OR_CONTROL_STORED_ENERGY
Block 1a prevention: ISOLATE_OR_CONTROL_STORED_ENERGY
Block 1a review reasons:

Narrative:
There was an opening in the bottom of the attached J-box to the load side of the adjacent breaker. This J-box was used as a raceway for cables going to the load side of the de-energized breaker and another set of cables that were abandoned. While disconnecting the drain wires from the cables one of the drain wires landed on the load side of the adjacent breaker and shorted out.

Block 1a cause evidence: While disconnecting the drain wires from the cables one of the drain wires landed on the load side of the adjacent breaker and shorted out.
Block 1a factor evidence: There was an opening in the bottom of the attached J-box to the load side of the adjacent breaker. This J-box was used as a raceway for cables going to the load side of the de-energized breaker and another set of cables that were abandoned.
Block 1a prevention rationale: Properly isolating and securing the cables would prevent the drain wire

# Corrected Factor-Prevention Alignment Examples - 6 of 6

from contacting the energized parts.

# Remaining Block 1a Review Cases - 1 of 5

These records still require review because of missing evidence, low support, nonacute routing, unusual prevention selection, or failed validation.

================================================================================
Example 1 | Record 220232500006 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: ENVIRONMENTAL_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
EE was working at the 24S welding and then went to help co-workers when EE was done. Upon return to the Shop employee began to feel nauseous and began vomiting. EE was taken to the ER to be seen. EE was treated for heat illness. EE was given a full release at the ER.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

================================================================================
Example 2 | Record 220252250002 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: ENVIRONMENTAL_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
Employee felt sick, shaking, weak. Treated at hospital for heat exhaustion.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

================================================================================
Example 3 | Record 220112940032 | No lost time or other
Block 1 cause: PRESSURE_OR_STORED_ENERGY_RELEASE
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: LINE_OF_FIRE_BODY_POSITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: ISOLATE_OR_CONTROL_STORED_ENERGY
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

# Remaining Block 1a Review Cases - 2 of 5

Narrative:
Using power washer. Wand on power washer got away from him. Sprayed hot water on his left leg.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

==========
Example 4 | Record 220100780003 | Permanent disability
Block 1 cause: CAUGHT_IN_OR_BETWEEN
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
The employee attempted to catch a Duke that was falling over. The duke pinched his finger against the face conveyor pan.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

==========
Example 5 | Record 220222510008 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: ENVIRONMENTAL_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
EE reports starting to feel sick around 2pm. AC unit in their truck blowing warm air since 11am. EE downed the truck at 2pm due to no AC. Seen by EMTs, and then at ER, diagnosed with heat exposure and dehydration, no treatment required, given one day off.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

==========
Example 6 | Record 220232420027 | Fatality
Block 1 cause: CAUGHT_IN_OR_BETWEEN
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL

# Remaining Block 1a Review Cases - 3 of 5

Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
While attempting to lower feed box on wash plant employee was trapped between feed  box and hand railing causing fatal injuries to employee.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

================================================================================
Example 7 | Record 220182130008 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
Working on the slope of leach pad walking up and down for about 3 hours. Employee felt dizzy and nauseous.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

================================================================================
Example 8 | Record 220222440019 | No lost time or other
Block 1 cause: OTHER
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: NOT_STATED
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: CONDUCT_PRE_JOB_HAZARD_REVIEW
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
Employee was operatring ATV all morning on the leach stockpile when they started feeling nauseous, light headed and weak.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

# Remaining Block 1a Review Cases - 4 of 5

```
================================================================================
Example 9 | Record 220071870019 | Days away
Block 1 cause: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
Employee using hammer and chisel while working on tail pulley in shop. Chisel slipped and hammer came
down on left thumb.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

================================================================================
Example 10 | Record 220073020011 | Fatality
Block 1 cause: STRUCK_BY_MOVING_OBJECT
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
Employee was working on a crusher. A bucket tooth was wedged in the crusher. The crusher was jacked up.
The jack kicked out and employee was struck in the head by an unknown object and knocked him into the
bed of the truck where he struck his head on an unknown object. Employee received head trauma.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

================================================================================
Example 11 | Record 220240640019 | Permanent disability
Block 1 cause: CAUGHT_IN_OR_BETWEEN
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
Changing out cylinder in JD 824 employee got tips of fingers between the mount and the cylinder cutting
```

# Remaining Block 1a Review Cases - 5 of 5

off tips of ring finger and middle finger.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

==========
Example 12 | Record 220090650024 | Permanent disability
Block 1 cause: STRUCK_BY_MOVING_OBJECT
Block 1a cause: INSUFFICIENT_INFORMATION
Block 1 factor: LINE_OF_FIRE_BODY_POSITION
Block 1a factor: NO_SEPARATE_FACTOR_STATED
Block 1 prevention: SECURE_OR_REPOSITION_MATERIAL
Block 1a prevention: INVESTIGATION_REQUIRED
Block 1a review reasons: Model response failed validation

Narrative:
While back flushing the emulsion system on the 14A stage loader, the employee was struck in the face by a hose.

Block 1a cause evidence:
Block 1a factor evidence:
Block 1a prevention rationale: The response could not be validated.

# Block 1a Freeze Decision Criteria

Recommended checks before freezing the system for Block 2:

1. Successful and schema-valid responses should remain high.

2. Review-required records should decline from the Block 1 level without hiding genuinely ambiguous records.

3. Factor-prevention alignment should improve substantially.

4. SECURE_OR_REPOSITION_MATERIAL should no longer dominate unrelated factors.

5. ESTABLISH_LINE_OF_FIRE_EXCLUSION should be used for supported line-of-fire and pinch-point cases.

6. USE_MECHANICAL_HANDLING_ASSISTANCE or REVISE_PROCEDURE_OR_JOB_PLAN should appear for supported manual-handling events.

7. Housekeeping should be limited to explicit walking, standing, access, or working-surface hazards.

8. Foreign-body, heat-exposure, acute strain, hernia, nonwork, and natural events should route consistently.

9. High-support examples should contain distinct evidence for both the event mechanism and the contributing factor.

10. The ontology, prompt, aliases, routing rules, compatibility map, and review rules should be frozen before Block 2.

# Technical Appendix

```
Model: Qwen/Qwen2.5-7B-Instruct
Prompt version: MSHA_BLOCK1A_ONTOLOGY_REFINEMENT_V2.0
GPU: Tesla T4
Block 1 development IDs requested: 1,000
Block 1a records found: 1,000
Missing development IDs: 0
Cause categories: 20
Factor categories: 17
Prevention categories: 18
Maximum retries: 3
Maximum new tokens: 600

Original Block 1 checkpoint:
/mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data Science/MSDS 453- AI
and Natural Language/Final Project/MSHA_Block1_Development_Checkpoint.jsonl
Block 1a checkpoint: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block1a_Refined_Checkpoint.jsonl
Block 1a comparison PDF: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW
Data Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block1a_Refinement_Comparison_Report.pdf
```